MCQs

1. B
2. B
3. C
4. A
5. B
6. B
7. C
8. C

No instructions were given for question 11 and 12 so I did in word file.

Question 11

Given: Parent node: 6 positives, 4 negatives (Total = 10 samples) After split:

- Left child: 5 positives, 1 negative (6 samples)
- Right child: 1 positive, 3 negatives (4 samples)

(a) Gini impurity of the parent node:

Gini = $1 - (p_1^2 + p_o^2)$

$p_1 = 6/10 = 0.6$, $p_o = 4/10 = 0.4$

Gini(Parent) = $1 - (0.6^2 + 0.4^2) = 1 - (0.36 + 0.16) = 1 - 0.52 =$ **0.48**

(b) Weighted Gini impurity of the children:

Left child: $p_1 = 5/6 \approx 0.833$, $p_o = 1/6 \approx 0.167$

Gini(Left) = $1 - (0.833^2 + 0.167^2) = 1 - 0.722 =$ **0.278**

Right child: $p_1 = 1/4 = 0.25$, $p_o = 3/4 = 0.75$

Gini(Right) = $1 - (0.25^2 + 0.75^2) = 1 - 0.625 =$ **0.375**

Weighted Gini = $(6/10)\times0.278 + (4/10)\times0.375 = 0.1668 + 0.15 =$ **0.3168**

(c) Gini Gain and is it a good split?

Gini Gain = Gini(Parent) - Weighted Gini(Children) = 0.48 - 0.3168 = **0.1632**

Yes, this is a good split because the Gini impurity decreased from 0.48 to 0.3168 after the split. The left child became much purer, so this split improves the decision tree.

Question 12

(a) Why is accuracy misleading here?

The dataset is highly imbalanced — fraud is only 1% of the data. If the model predicts "not fraud" for every transaction, it will be correct 99% of the time because 99% of transactions are actually not fraud.

So even though accuracy is 99%, the model is useless because it fails to detect any fraud cases.

(b) Two concrete remedies:

1. **Use Class Weighting**: Give higher weight to the fraud class while training. This forces the model to pay more attention to the rare fraud cases.
2. **Use F1-Score or Recall instead of Accuracy**: These metrics focus on how well the model detects actual fraud, not overall correctness.

Question 9

(a) Compute the mean of x and y:-

Mean of $x(\bar{x}) = 1+2+3+4 / 4 = 2.5$

- - $y(\bar{y}) = 2+2+4+5 / 4 = 3.25$

(b) Using closed form formula:-

Formula:- $\frac{\Sigma(x_i-\bar{x})(y_i-\bar{y})}{\Sigma(x_i-\bar{x})^2} = w$

x	y	$x-\bar{x}$	$y-\bar{y}$	$(x-\bar{x})(y-\bar{y})$	$(x-\bar{x})^2$
1	2	-1.5	-1.25	1.875	2.25
2	2	-0.5	-1.25	0.625	0.25
3	4	0.5	0.75	0.375	0.25
4	5	1.5	1.75	2.625	2.25

sum

Numerator $= 5.5$

Denominator $= 5$ sum

$w = 5.5/5 = 1.1$

Intercept,

$b = \bar{y} - w \times \bar{x}$
$= 3.25 - 1.1 \times 2.5$
$= 0.5$

Equation:-

$y = 1.1x + 0.5$

(c) Predicted val for $x = 5$ point (3,4)

Predicted val for $x = 5$

$y = 1.1 \times 5 + 0.5$
$= 6$

Residual for point (3,4)
Residual $= y - \hat{y} = 4 - (1.1 \times 3 + 0.5)$
$= 4 - 3.8$
$= 0.2$

Predicted val $= 5 \rightarrow 6.0$
Residual for point $= (3,4) = 0.2$

Date

Question 10

Predicted Probabilities : $P(y=1|x) = [0.9, 0.4, 0.8, 0.3, 0.6]$

True labels : y_true = [1, 0, 0, 1, 1]

(a) Step 1 :- Converting probabilities to binary predictions using threshold = 0.5

Test Point	Probability	Predicted	True Label
1	0.9	1	1
2	0.4	0	0
3	0.8	1	0
4	0.3	0	1
5	0.6	1	1

Step 2 :- Confusion Matrix (Class 1 Pos)

- True Positive = 2 (1 and 5)
- FP = 1 (3)
- FN = 1 (4)
- TN = 1 (2)

Predicted Label

1 if prob ≥ 0.5

0 if < 0.5

Signature

No.

Calculating Metrics

$$\text{Accuracy} = \frac{TP + TN}{\text{Total Samples}} = \frac{2+3}{5} = 0.60 \text{ or } 60\%$$

Precision for Class 1

$$= \frac{TP}{TP + FP} = \frac{2}{2+3} = 0.667 = 66.7\%$$

Recall For Class 1

$$= \frac{TP}{TP + FN} = \frac{2}{2+1} = 0.667 \text{ or } 66.7\%$$

$$\text{F1-Score} = \frac{2 \times (\text{Precision} \times \text{Recall})}{\text{Precision} + \text{Recall}}$$

$$= \frac{2 \times (0.667 \times 0.667)}{0.667 + 0.667} \Rightarrow 0.667 \Rightarrow 66.7\%$$

Date

(b) Scenario

In malaria detection system we prefer a threshold of 0.3 over 0.5.

Justification

If the model misses a patient who actually has malaria (FN) it can be life threatening b/c the patient will not get treatment.

Even though lowering the threshold may give unessecy medicine to healthy patients but it is better to catch real patients than to miss them. Doctors can do further tests later to confirm